

Measuring Technological Innovation over the Long Run

Bryan Kelly, Dimitris Papanikolaou, Amit Seru, and Matt Taddy

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Motivation & Research Question

- Much of long-run growth is attributed to technological progress — but it is hard to measure consistently over time.
- Patent counts treat all patents as equal; forward citations are only consistently recorded after 1947, are discrete, and depend on examiner/inventor discretion.
- Goal: build indicators of patent importance from patent TEXT alone, so they are available for the entire history of US patents (1840–2010).
- Key idea: an important patent is both NOVEL (unlike prior art) and IMPACTFUL (similar to future work).

Data

- Full text of 9+ million US patents, 1840–2010, from the USPTO and Google's patent search engine.
- Unstructured patent text converted into numerical (bag-of-words) representations.
- Post-1947 citation data and Kogan et al. (2017, KPSS) market-value estimates used only for external validation.
- No inputs required besides the patent document text — the source of the method's long historical reach.

NLP Method (1): Text Similarity between Patents

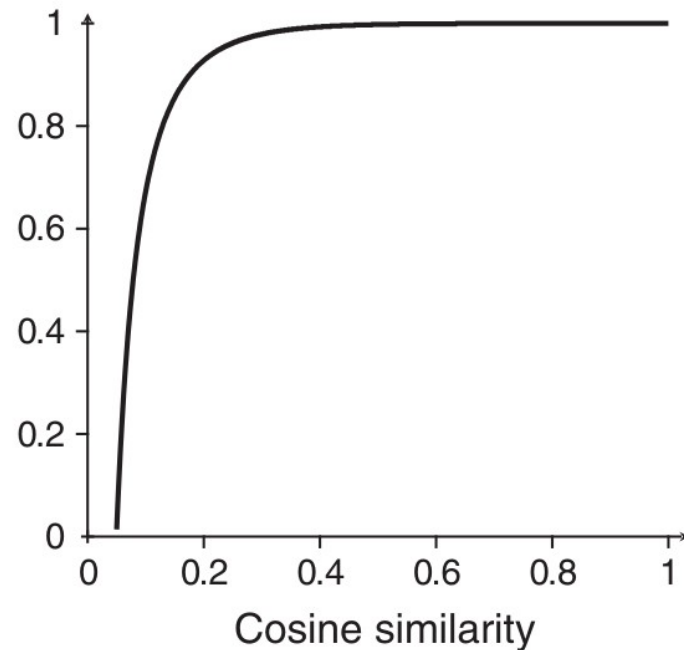
- Represent each patent as a weighted word-count vector using TF-IDF (term frequency $\times$ inverse document frequency).
 - Problem with standard IDF: it down-weights terms that later became common (e.g. 'alternating current'), inverting a pioneering patent's importance.
- Innovation — Backward-IDF (BIDF): weight terms by how rare they were among patents granted BEFORE the focal patent, using the earlier patent's year for each pair.
- Pairwise similarity $p(i,j)$ = cosine similarity of the two normalized TFBIDF vectors; $p \in [0,1]$.

NLP Method (2): From Similarity to an Importance Score

- Backward similarity BS_j = sum of similarities to patents in the prior 5 years → low BS means NOVEL.
- Forward similarity FS_j = sum of similarities to patents in the next 10 years → high FS means IMPACTFUL.
- Patent importance: $q_j = FS_j / BS_j$ — high for patents that break from prior art yet shape future work.
- 'Breakthrough' patents = top 10% of importance (net of issue-year fixed effects); indices = breakthroughs per capita per year.

Similarity Is Validated by Citation Linkages

Panel A. Empirical CDF



Panel B. Probability of citation pair

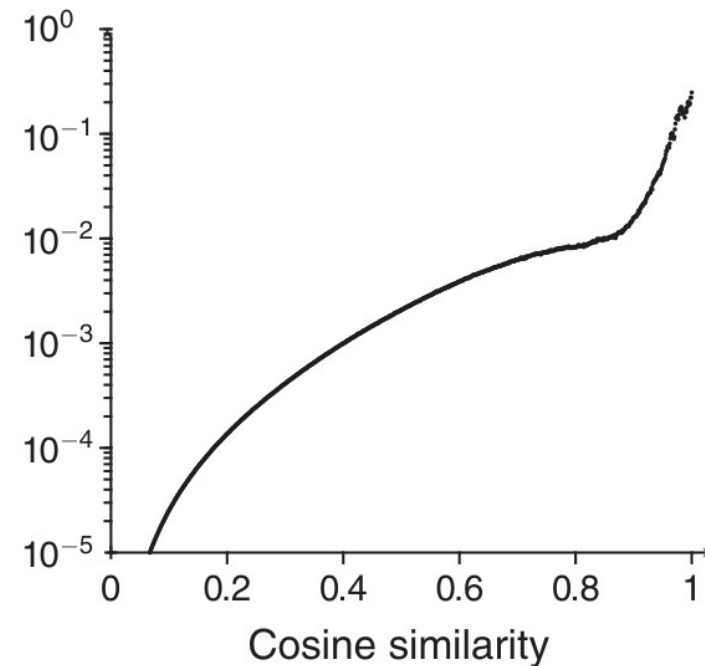


FIGURE 1. PAIRWISE SIMILARITY AND CITATION LINKAGES

Note: Panel A: empirical CDF of the text similarity score $p(i,j)$ across patent pairs (most pairs are highly dissimilar; median $\approx 7.8\%$). Panel B: the probability that patent j cites earlier patent i rises monotonically with their text similarity. Post-1945 data. (Figure 1)

Importance Predicts Citations and Market Value

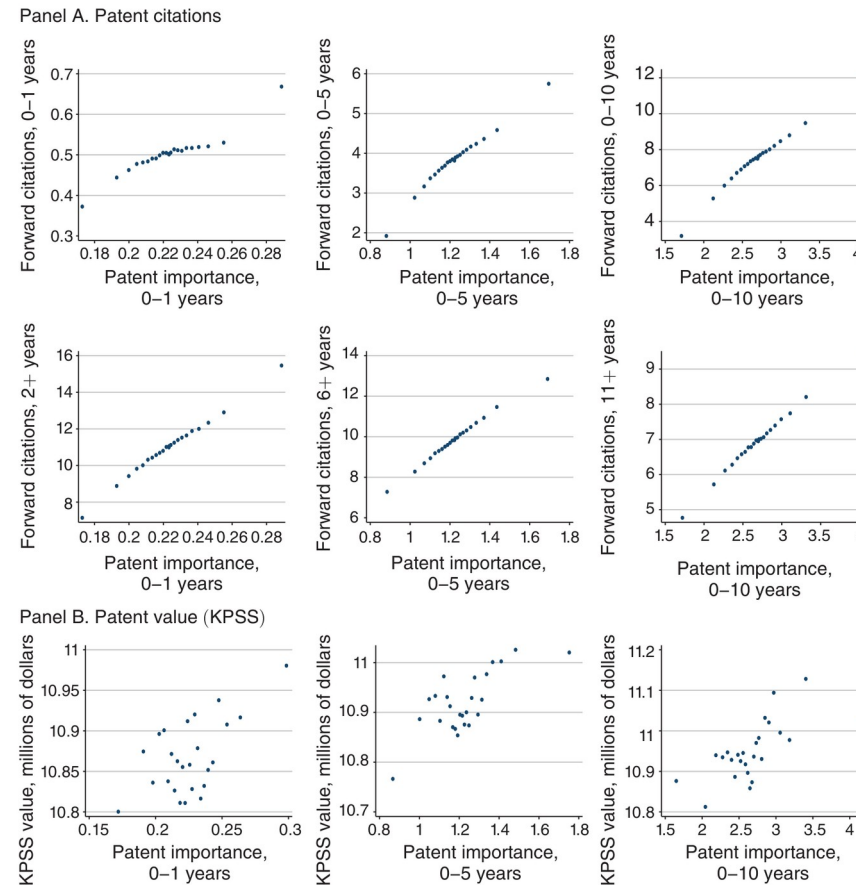


FIGURE 3. PATENT IMPORTANCE: VALIDATION

Note: Binned scatter plots relating the text-based importance measure to forward citations (Panel A) and to KPSS stock-market patent value (Panel B), controlling for technology class and assignee×grant-year fixed effects. Importance is contemporaneously correlated with citations AND predicts future citations beyond same-horizon counts. (Figure 3)

Headline Result: Three Long-Run Innovation Waves

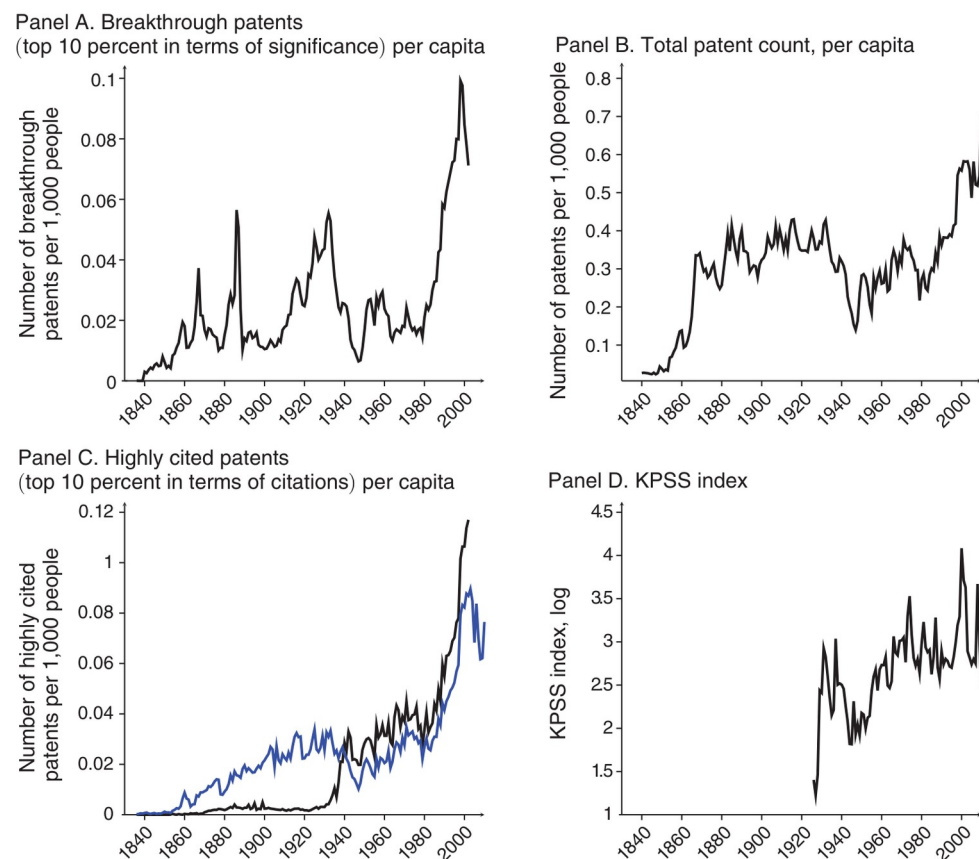
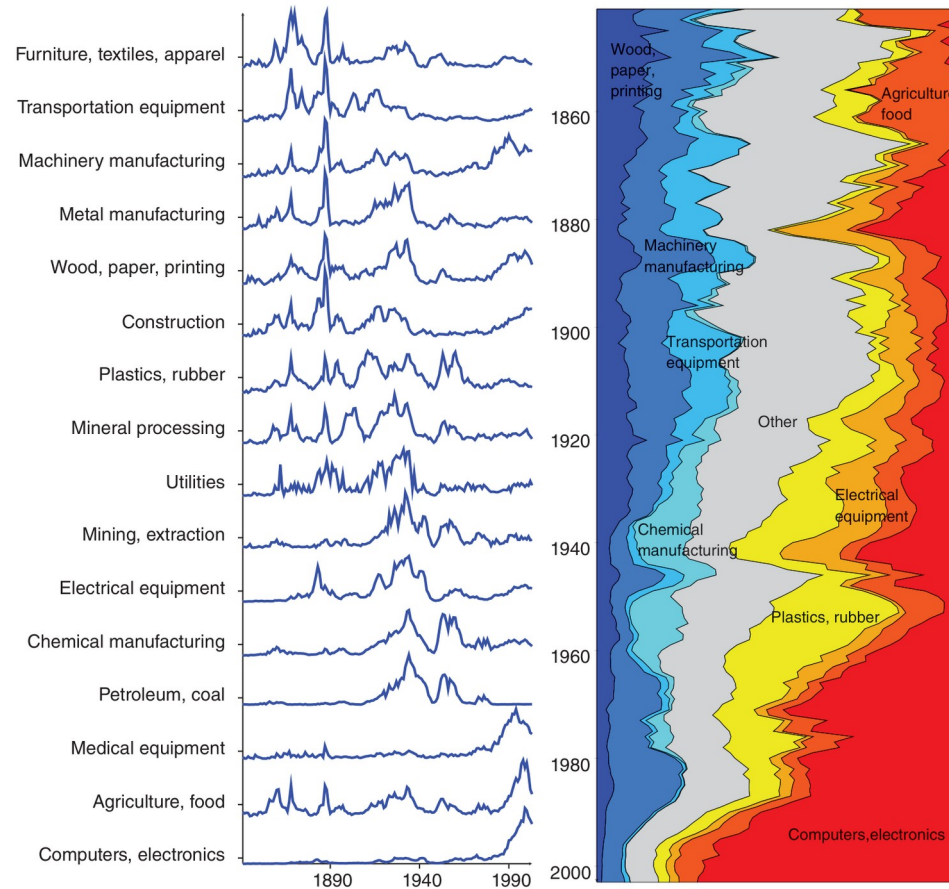


FIGURE 4. TECHNOLOGICAL INNOVATION OVER THE LONG RUN: NEW VERSUS EXISTING INDICATORS

Note: Panel A: breakthrough patents per capita (new text-based index) reveals waves ~1870–80, 1920–35, and 1985–present. Panels B–D: existing indicators — raw patent counts, citation-based index, and KPSS value index — miss pre-1940 innovation because of data limitations. (Figure 4)

Sectoral Indices: Shifting Sources of Breakthroughs



Note: Per-capita breakthrough patents by industry (3-digit NAICS). Breakthroughs shift from engineering/construction/textiles (1840–70), to electricity and transportation (from ~1870), to chemicals/plastics (1920s–30s), to computers & electronics (post-1980). (Figure 5)

Contributions & Takeaways

- A text-only measure of patent importance that is consistently available back to 1840 — unlike citation- or market-based measures.
- Validated three ways: historically significant patents score high (mean rank ~0.74–0.78), correlates with and predicts citations, and correlates with market value.
- Reveals long-run technological waves and their changing sectoral origins, linked to productivity growth.
- Method blueprint: TF-IDF + backward-IDF + cosine similarity is a general recipe for measuring novelty and impact in any evolving text corpus.